

PaperBridge

Software Requirements Specification

Development Review and Quotation Package

1. Executive Summary

PaperBridge is a cloud-based, AI-powered advocacy and care coordination platform designed to help families organize, understand, and securely share information related to children and dependents receiving educational, medical, therapeutic, or developmental support.

The platform combines document management, OCR processing, AI-assisted analysis, secure sharing, calendar management, notifications, and family collaboration into a unified web and mobile experience. The objective is to provide families with a centralized source of truth while maintaining complete control over privacy, security, and access permissions.

This document is intended to provide sufficient information for a senior software architect or development team to evaluate the project, review the proposed architecture, identify risks, and prepare a development quotation.

2. Business Objective

Families often manage records across multiple providers, schools, therapists, insurance companies, and healthcare systems. Important documents become scattered across emails, portals, paper files, and local storage, making it difficult to locate information when needed.

PaperBridge is intended to solve this problem by providing a centralized platform where users can:

- Store and organize documents
- Search across all records
- Generate AI summaries of complex reports
- Prepare for appointments and meetings
- Share information securely with providers and caregivers
- Track appointments and important events
- Maintain a complete history of activity and access

The long-term goal is to become a comprehensive family advocacy platform rather than simply a document repository.

3. System Workflow

The following workflow represents the primary user journey envisioned for the PaperBridge platform.

Family Onboarding

1. A Family Unit Administrator creates a PaperBridge account.
2. The administrator creates profiles for children, dependents, or other family members.
3. Additional family members may be invited into the Family Unit with configurable permissions.
4. The administrator configures security settings and subscription preferences.

Document Management Workflow

1. Users upload documents into the appropriate profile.
2. Documents are categorized manually or through AI-assisted categorization.
3. OCR processing extracts searchable text from supported documents.
4. AI generates summaries, recommendations, and key insights.
5. Documents become searchable through metadata and OCR content.
6. Documents may be organized, tagged, shared, or archived.

Collaboration Workflow

1. Family members access assigned profiles and documents.
2. External providers may receive secure document access links.
3. Access activity is recorded within the audit log.
4. Administrators may revoke access at any time.

Care Coordination Workflow

1. Users create appointments, meetings, and reminders.
2. Events may be associated with profiles and supporting documents.
3. Notifications are generated based on event schedules.
4. AI-generated insights may be reviewed before appointments and meetings.

4. User Types

The system will support three primary user roles.

Family Unit Administrator

The Family Unit Administrator serves as the primary account owner and controls all account settings, billing, profiles, permissions, and sharing activities.

Responsibilities include:

- Managing subscription and billing
- Creating and maintaining family profiles
- Inviting family members
- Assigning permissions
- Managing external sharing
- Reviewing audit activity

Family Unit Profile User

These users are invited into the platform by the administrator and may include spouses, grandparents, caregivers, or adult dependents.

Capabilities are controlled by the administrator and may include:

- Viewing documents
- Uploading documents
- Viewing AI summaries
- Managing calendar events
- Accessing assigned profiles

External Viewer

External Viewers do not maintain platform accounts.

Examples include:

- Teachers
- Therapists
- Physicians
- Attorneys
- Care Coordinators

Access is limited to:

- Specifically shared documents
- Time-limited secure links
- Read-only access

5. Platform Requirements

PaperBridge will consist of a web application, mobile applications, cloud infrastructure, document storage services, AI services, and supporting administration tools.

Web Application

The web application will serve as the primary administrative interface and must support:

- Responsive layouts
- Desktop and tablet usage
- Modern browser compatibility
- Full administrative functionality

Supported browsers include Chrome, Edge, Safari, and Firefox.

Mobile Applications

Native-quality mobile applications will be provided for iOS and Android.

Primary mobile functions include:

- Document capture
- Document viewing
- AI summary review
- Notifications
- Calendar access
- Secure sharing

6. Authentication & Identity Management

The platform must provide secure authentication and account management.

Required capabilities include:

- Email/password login
- Email verification
- Password reset
- Invitation-based user onboarding
- Session management
- Device management
- Multi-factor authentication

Multi-factor authentication will be mandatory for Family Unit Administrators and optional for Profile Users.

Future support should be considered for:

- Google Authentication
- Apple Authentication
- Microsoft Authentication
- Enterprise SSO

7. Family Profile Management

Each family account may contain multiple managed profiles.

Examples include:

- Children
- Parents
- Grandparents
- Dependents
- Care recipients

The system must allow users to:

- Create profiles
- Edit profile information
- Archive profiles
- Assign permissions
- Associate documents and calendar events

Profiles serve as the primary organizational structure throughout the application.

8. Document Categories

PaperBridge will support document categorization to improve organization, permissions management, search capabilities, and AI processing.

Medical

- Medical Records
- Physician Notes
- Diagnostic Reports
- Laboratory Results
- Medication Information

Therapy

- Speech Therapy Reports
- Occupational Therapy Reports
- Physical Therapy Reports
- Behavioral Therapy Documentation

Educational

- IEP Documents
- School Evaluations
- Progress Reports
- Academic Records
- Teacher Communications

Insurance

- Insurance Claims
- Explanation of Benefits
- Coverage Documentation
- Authorizations

Legal

- Guardianship Documents
- Legal Agreements
- Advocacy Documentation

General

- Personal Notes
- Meeting Notes
- Miscellaneous Documents

Additional categories should be configurable through future administrative functionality.

9. Document Management

Document management is the core feature of the platform.

Supported File Types

Documents

- PDF
- DOC/DOCX
- TXT
- RTF

Images

- JPG/JPEG
- PNG
- TIFF
- HEIC

Spreadsheet Files

- XLS/XLSX
- CSV

Document Functions

- Upload
- Download
- Preview
- Categorization
- Metadata Editing
- Version History
- Soft Delete
- Restore
- Bulk Actions
- Export

Document Organization

Documents must be associated with:

- Profiles
- Categories
- Tags
- Metadata

Version history should preserve previous document revisions and associated audit records.

10. Artificial Intelligence

Artificial intelligence is intended to help users understand information rather than make decisions for them.

AI Objectives

The AI system should assist users in understanding complex information, preparing for important meetings, identifying key recommendations, and organizing information more effectively.

AI Services

Document Summarization

Generate:

- Executive Summaries
- Key Findings
- Recommendations
- Progress Highlights
- Suggested Next Steps

AI Categorization

Suggest document classifications based on content analysis.

Metadata Extraction

Extract relevant dates, providers, organizations, diagnoses, recommendations, and other important information.

Advocacy Assistance

Provide support for:

- IEP Preparation
- Appointment Preparation
- Therapy Reviews
- Progress Reviews
- Follow-Up Recommendations
- Missing Information Identification

AI Requirements

- User-editable summaries
- Version tracking
- Audit logging
- Human review of all outputs

The preferred AI provider is Azure OpenAI.

Customer data must not be used for model training.

Future AI Enhancements

The architecture should support:

- Semantic Search
- Retrieval-Augmented Generation (RAG)
- Cross-Document Analysis
- Timeline Generation
- Meeting Preparation Briefs
- Care Coordination Recommendations

All AI-generated outputs must remain reviewable, editable, and auditable.

11. OCR & Document Processing

PaperBridge must support OCR processing for scanned and image-based documents.

Document Workflow

Upload → OCR Processing → Searchable Text → AI Analysis → Search Indexing

Required capabilities include:

- OCR extraction
- Processing status monitoring
- Retry processing
- Error reporting
- Searchable text storage
- OCR audit logging

The OCR engine should be architected to support future improvements and scaling.

12. Search & Information Retrieval

The platform must support powerful search capabilities across both document metadata and OCR-generated content.

Initial search functionality should include:

- Document search
- Profile search
- OCR text search
- Filtering by category

- Filtering by date
- Filtering by profile

Future enhancements may include:

- Semantic search
- Vector search
- Natural language search
- Retrieval-Augmented Generation (RAG)

13. Role-Based Access Control

Access to information must be controlled through a granular permissions model.

Permissions should be assignable at:

- Profile level
- Category level
- Document level
- Calendar level

Supported actions include:

- View
- Upload
- Edit
- Delete
- Share
- Revoke
- Export

All permission changes must be audit logged.

14. Secure Sharing & Collaboration

Secure sharing is a critical feature of the platform.

Internal Sharing

Allows authenticated users within the family unit to access information according to assigned permissions.

External Sharing

Allows secure distribution of documents using tokenized links.

Supported controls include:

- Link expiration
- Password protection
- Download restrictions
- Access revocation
- Activity tracking

Future enhancements may include:

- Comments
- Annotations
- Collaborative workflows

15. Calendar & Care Coordination

The platform will include a calendar and reminder system for managing appointments, meetings, evaluations, therapy sessions, and deadlines.

Capabilities include:

- Event creation
- Event editing
- Reminders
- Notifications
- Event categorization
- Profile associations

Future integrations may include:

- Google Calendar
- Microsoft Outlook
- ICS Feeds

16. Notifications

Users should receive notifications regarding important platform activity.

Initial channels include:

- Email
- In-Application Notifications

Notification categories include:

- Shared documents
- Viewed documents
- Calendar reminders
- Security events
- Account changes

Future enhancements may include:

- Push Notifications
- SMS Messaging

17. Subscription & Billing

PaperBridge will operate as a subscription-based SaaS platform.

Billing functionality should support:

- Subscription plans
- Upgrades
- Downgrades
- Cancellation
- Invoice history
- Payment methods
- Usage tracking

Stripe is the preferred payment processor.

Subscription Model

Family Plan

- Up to 5 Profiles
- Annual Billing

Family Plus Plan

- Up to 10 Profiles
- Annual Billing

Family Unlimited Plan

- More Than 10 Profiles
- Annual Billing

Usage Metrics

The platform should track:

- Number of Profiles
- Storage Consumption
- OCR Usage
- AI Usage
- Shared Documents
- External Viewers

18. Data Retention & Storage Management

PaperBridge must provide policies governing data retention, recovery, and storage management.

Document Retention

- Active documents retained indefinitely
- Soft-deleted documents retained for 90 days
- Version history retained while documents remain active

Audit Retention

- Audit logs retained for seven years

Backup Retention

- System backups retained for a minimum of 30 days

File Management

- Configurable upload limits
- Virus and malware scanning
- Secure object storage
- Version tracking

19. Security & Compliance

Security is a foundational requirement.

Requirements include:

- U.S.-based data residency
- TLS 1.2+ encryption in transit
- AES-256 encryption at rest
- Document-level encryption
- MFA support
- Secure session management
- Role-based access controls
- Audit logging

The architecture should support future HIPAA and SOC 2 initiatives if required.

20. Audit Logging

The platform must maintain a complete audit trail of system activity.

Events to capture include:

- Authentication events
- Permission changes
- Document access
- Document sharing
- AI activity
- Calendar activity
- Billing activity

Audit logs should:

- Be retained for seven years
- Support CSV export
- Support JSON export
- Be searchable and filterable

21. Reporting & Analytics

The platform should provide administrative and operational reporting.

Reporting capabilities should include:

- Document counts
- Profile counts
- Storage usage
- OCR usage
- AI usage
- Sharing activity
- User activity
- Audit reporting

Reports should support filtering and export capabilities.

22. System Integrations

Initial Integrations

- Azure OpenAI
- Stripe

Future Integrations

- Google Calendar
- Microsoft Outlook
- School Systems
- EMR/EHR Platforms
- Therapy Platforms
- Insurance Providers

The architecture should support future integration through secure APIs and extensible service layers.

23. Scalability & Infrastructure

The platform should initially support:

- 1,000 Family Units
- 5,000 Profiles
- 125,000 Documents
- 2,000 Users
- 250 Concurrent Users

The architecture should scale without significant redesign to support future growth into tens of thousands of families and millions of documents.

Current preferred technologies include:

- Angular
- Flutter
- Python / FastAPI
- PostgreSQL
- Azure Blob Storage
- Azure OpenAI
- Qdrant or ChromaDB
- Microsoft Azure

Development teams may recommend alternative technologies if they provide meaningful advantages in scalability, maintainability, cost, or development efficiency.

24. Non-Functional Requirements

Performance

- User authentication should complete within acceptable user experience thresholds.
- Search results should return within a few seconds.
- Document uploads should provide immediate user feedback.
- AI processing should provide status visibility.

Availability

- Target availability of 99.9% uptime excluding planned maintenance.

Scalability

- Support growth without architectural redesign.
- Support horizontal scaling where appropriate.

Security

- Zero-trust security principles.
- Multi-factor authentication.
- Encryption of sensitive information.

Maintainability

- Documented APIs.
- Automated deployment pipelines.
- Centralized logging and monitoring.

25. Disaster Recovery & Business Continuity

The platform should include disaster recovery procedures and business continuity planning.

Backup Strategy

- Daily backups
- Automated backup validation
- Secure offsite storage

Recovery Objectives

- Recovery Time Objective (RTO): 24 Hours
- Recovery Point Objective (RPO): 1 Hour

Infrastructure should be designed to minimize downtime and data loss.

26. Project Assumptions

The following assumptions have been made for the initial MVP:

- All customer data remains within the United States.
- English is the only supported language.
- Azure is the preferred cloud hosting platform.
- Azure OpenAI is the preferred AI provider.
- Stripe will be used for payment processing.
- HIPAA certification is not required for the MVP.
- Mobile and web applications will use a shared backend.
- External Viewers will not maintain persistent platform accounts.
- Customer data will not be used for AI model training.

Changes to these assumptions may impact scope, timeline, architecture, and cost.

27. MVP Success Criteria

The MVP shall be considered complete when the following capabilities are operational.

User Management

- Family Unit accounts can be created.
- Family profiles can be managed.
- Family members can be invited and assigned permissions.

Document Management

- Documents can be uploaded and organized.
- OCR processing is operational.
- Documents are searchable.

Artificial Intelligence

- AI summaries can be generated.
- AI categorization is operational.

Sharing

- Documents can be securely shared internally and externally.
- Access can be revoked.
- Sharing activity is audited.

Care Coordination

- Calendar events can be created and managed.
- Notifications are operational.

Security & Compliance

- Audit logging is functional.
- Encryption requirements are implemented.
- Role-based permissions are enforced.

Billing

- Subscription management and payment processing are operational.

Mobile Access

- iOS and Android applications are available and connected to the shared backend platform.

28. Conclusion

PaperBridge is a secure, AI-powered family advocacy platform that combines document management, OCR processing, AI-assisted analysis, secure sharing, care coordination, auditability, and subscription management into a single scalable solution.

The purpose of this specification is to provide sufficient technical and business context for architecture review, feasibility assessment, risk analysis, and development estimation while allowing development teams flexibility in proposing the most appropriate technical implementation.